

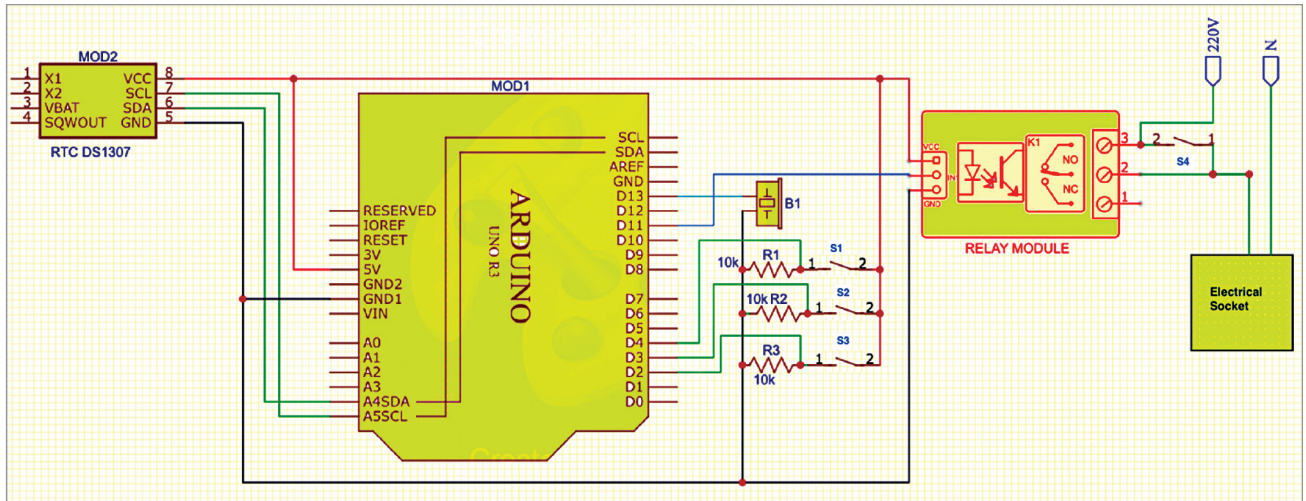


Fig. 2: Circuit diagram

and keep switches S1 and S3 open.

Mode 4: Office mode. In this mode it operates continuously during office hours from 10:00am to 6:00pm, which is suitable for open office rooms. To enable this mode, close switches S1 and S2 and keep switch S3 open.

Mode 5: Office toggle mode. The repellent switches on and off hourly during office hours from 10:00am to 6:00pm, which is suitable for closed, air-conditioned office rooms. To enable this mode, close switch S3 and keep switches S1 and S2 open.

Mode 6: Hall mode. This mode is for use in halls during evening gatherings to keep the repellent switched on from 5:00pm to 10:00pm. To enable this mode, close switches S1 and S3 and keep switch S2 open.

Mode 7: All day toggle mode. This mode operates the repellent continuously but non-sequentially, switching on and off every hour, which is ideal for office rooms operating 24/7. To enable this mode, keep switch S1 open and switches S2 and S3 closed.

The seven modes are controlled by push-to-on switches S1, S2, and S3, as shown in Fig. 1. These switches (S1 through S3) are simple

Arduino Uno Pins	Components such as RTC module, switches, buzzer etc
5V pin	DS1307 I2C RTC module VCC pin
5V pin	2nd pins of S1, S2, S3
5V pin	Connected to VCC pin of relay module
Pin D2	1st pin of S3
Pin D3	1st pin of S2
Pin D4	1st pin of S1
A4	SDA pin of DS1307 I2C RTC module
A5	SCL pin of DS1307 I2C RTC module
Pin D11	Input signal pin of relay module
Pin D13	Buzzer positive pin
GND	GND pin of relay module
GND	One end of resistors R1, R2, R3
GND	GND pin of buzzer
GND	GND pin of DS1307 I2C RTC module

on/off switches connected in series with the resistor between the 5V and the ground pin of the Arduino board.

Green boxes in Table 3 indicate the on-time of the liquid mosquito repellent. These switches are connected to the Arduino board pins as shown in Table 2. The positions of the switches and the corresponding modes of operation are shown in Table 3.



Fig. 3: DS1307 I2C RTC module

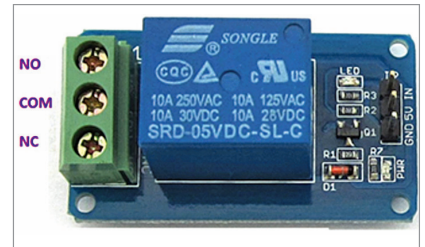


Fig. 4: One-channel 5V relay module

Software

The circuit operation is governed by the software program loaded into the internal memory of the Arduino Uno. The program, written in Arduino programming language, is uploaded using the Arduino IDE. The sketch initialises input and output pins and utilises the wire library for communication with the DS1307 RTC module. Serial communication